# RYAN AHRARI

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## SUMMARY


Software Engineer with real-world, production experience across full-stack development, distributed systems, and enterprise software delivery, combined with hands-on experience evaluating AI-generated code and training large language models on software engineering tasks. Skilled at reviewing code for correctness, performance, and clarity, and communicating structured, actionable feedback to both technical and non-technical audiences. Proficient in Python, Java, C++, TypeScript, and SQL, with applied machine learning and AI-evaluation expertise (TensorFlow, model interpretability, explainability methods) and practical fluency in AI-assisted development workflows (Claude, OpenAI).


## TECHNICAL SKILLS

**Languages:** Python | JavaScript | TypeScript | Java | C | C++ | PL/SQL | SystemVerilog | Assembly

**AI Dev Tools & Evaluation:** Claude | OpenAI | AI Code Review | LLM Training Data | Data Annotation

**ML & AI:** TensorFlow | Keras | scikit-learn | CNNs | Transformers | Transfer Learning | Computer Vision | Flair NLP | MATLAB | MLOps | Predictive Analytics | Deep Learning Recommendation Systems | AI Ethics & Governance

**Web:** React | Next.js | HTML | CSS | REST APIs | Responsive UI

**Data & Databases:** SQL | NoSQL | Firebase | Oracle E-Business Suite (EBS) | Relational Databases

**Systems & Infrastructure:** Linux | Kubernetes | Docker | CI/CD Concepts | Computer Architecture | Operating Systems | Parallel Programming

**Tools & Practices:** Git | Agile | SDLC | RBAC | Functional Testing | API Integration

## WORK EXPERIENCE

**AI Trainer** | DataAnnotation (Remote) *Feb 2024 – Present*

- Design and evaluate coding problems across Python, JavaScript, React, and C++, providing structured feedback to train large language models on software engineering tasks.
- Review AI-generated code for correctness, performance, and clarity, helping improve model understanding of programming tasks, system design, and engineering best practices.
- Apply structured testing and documentation practices consistent with SDLC quality standards, building hands-on expertise with AI-assisted development tools including OpenAI and Claude.

**Software Associate** | Infosys Limited (Remote) *Jun 2022 – Sep 2023*

- Designed and implemented high-performance SQL queries and PL/SQL procedures for Oracle E-Business Suite, applying algorithmic and query-optimization techniques to streamline data processing for enterprise clients.
- Resolved critical database integrity issues and automated manual validation workflows by writing scripts and tools, ensuring SLA compliance while reducing repetitive engineering tasks.
- Partnered with stakeholders and cross-functional teams on requirements gathering, solution documentation, and debugging of enterprise-scale applications using Agile methodologies.

**Computer Science Tutor** | Baskin School of Engineering, UC Santa Cruz *Oct 2019 – Jun 2022*

- Tutored 100+ undergraduate students in data structures, algorithms, systems programming, and computer architecture, improving exam pass rates by 20%.
- Delivered lectures on logic design applications, including FPGA prototyping and hardware description languages.
- Communicated complex technical concepts clearly, adapting explanations to individual student needs — developing strong structured-feedback and presentation skills.

## PROJECTS

**MetalsMarket** | Full-Stack Marketplace App | Solo Developer *Dec 2025 – Present*

- Build a full-stack marketplace platform for buying and selling metals, developed end-to-end as sole developer using React, Next.js, TypeScript, and SQL.

- Implement secure user authentication with role-based access control (RBAC) and a relational database schema with advanced search and filtering by metal type, grade, quantity, and price range.
- Apply enterprise development practices including modular architecture, Git-based version control, and iterative deployment planning; integrating Stripe for payment processing.

**Malaria Detection with Deep Learning** | Project *Apr 2025 – Jun 2025*

- Developed a CNN and transfer-learning model to automate malaria detection from blood cell images, achieving 95%+ accuracy.
- Applied Python, TensorFlow, and data augmentation to address real-world challenges in medical image classification, reducing manual diagnosis time.

**Street View Housing Number Digit Recognition** | Project *Mar 2025 – Apr 2025*

- Trained CNNs and ANNs on the SVHN dataset to automate digit recognition for applications such as map-based geolocation.
- Compared architectures in TensorFlow/Keras, achieving high-accuracy digit prediction from complex natural scenes.

**FoodHub Order Analysis** | Project *Feb 2025 – Mar 2025*

- Performed exploratory data analysis on 10K+ food delivery orders to uncover trends in cuisine demand, peak ordering times, and delivery efficiency.
- Applied statistical analysis and visualization to generate data-driven insights that improved restaurant partnerships and operational workflows.

**AI Explainability & Interpretability Analysis** | UC Santa Cruz *Jan 2023 – Jun 2023*

- Built and benchmarked NLP sentiment-analysis pipelines with Flair, comparing SHAP, LIME, and attention-mechanism explainability methods across AI research papers to evaluate trends in Explainable AI (XAI) and model interpretability.
- Published findings as a master's thesis, analyzing implications for ethical AI governance and responsible AI systems.

**Real-Time Reaction Game (FPGA/ASIC)** | UC Santa Cruz *Jan 2023 – Mar 2023*

- Built a real-time reaction game on an iCEBreaker FPGA in SystemVerilog, implementing FSM-based control logic and LFSR-based randomization.
- Validated timing and functionality via RTL simulation (Icarus Verilog/Verilator) before synthesizing for low-power FPGA deployment.

**Kubernetes Resiliency Protocol** | Dell *Jan 2022 – Jun 2022*

- Built a Python/REST API-based resiliency protocol for Kubernetes, improving cluster uptime by 90% through etcd metadata optimization.
- Simulated node failures and network partitions to stress-test fault tolerance, documenting root causes and remediation steps.

## EDUCATION

**M.S. Computer Science and Engineering** | University of California, Santa Cruz *2023*
Thesis: Explainability and Interpretability Analysis in AI
Coursework: Machine Learning Theory | Computer Vision | AI Ethics & Governance

**B.S. Computer Engineering** | University of California, Santa Cruz *2022*
Focus: Systems Programming, Computer Architecture, and Embedded AI
Coursework: Computer Architecture | ASIC/FPGA Design | Operating Systems | Data Structures & Algorithms

**Applied Data Science Program (Certificate)** | MIT *2025*
Coursework: Large-Scale Recommendation Systems | Deep Learning for Computer Vision

## CERTIFICATIONS

AZ-900 Microsoft Certified Azure Fundamentals
Google Advanced Generative AI for Developers